Supplementary material for

# Challenging One-Way Fisheries Modelling: Two-Way Coupling Reveals Critical Marine Ecosystem Dynamics

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## S1 FABM-MIZER Model description

In the section below, the discretised FABM-MIZER community size spectrum model is described: equations are given for individual size classes. Parameters primarily taken from Blanchard et al. (2009, 2012, 2014) can be found in Table S1. Physiological rates in the model are assumed to be temperature dependent (section S1.7). Temperature dependence is applied to predator-prey interactions (thus growth and predation mortality vary with temperature) and background mortality, but not senescence mortality.

### *S1.1 Discretization of the size spectrum*

Size-structured models are primarily formulated in continuous-size space (Blanchard et al., 2014, 2009; Hartvig et al., 2011) producing an advection-reaction-type partial differential equation that describes the evolution of total abundance as a function of individual wet mass:

$$\frac{\partial}{\partial t}N(w) = \frac{\partial}{\partial w}(gN) - \mu(w) \qquad (S1)$$

Here, $N$ denotes abundance (#), $w$ individual wet mass (g), and $g$ individual growth (g time$^{-1}$) with $\mu(w)$ indicating various loss terms associated with predation, natural mortality and fishing. In practice equation S1 is usually solved by first discretizing individual mass with a first-order upstream scheme, resulting in a model for the evaluation of abundance per size class, $N(w_i)$. Once discretised, equation S1 becomes

$$\frac{\partial}{\partial t}N(w_i) = \frac{-g(w_i)N(w_i) + g(w_{i-1})N(w_{i-1})}{w_i - w_{i-1}} - \mu(w_i) \qquad (S2)$$

Where $g(w_i)$ is the instantaneous individual growth rate in size class $i$ (g time$^{-1}$).

To incorporate the community size-based fish model with FABM, the source terms for each size class of fish, as well as prey and waste, are computed, and provided as-is to the hydrodynamic model to be time-integrated using the same solver it uses for all other model variables.

### *S1.2 Vertical distribution*

The vertical distribution of fish is assumed to be primarily the result of active swimming by the

fish themselves and not the result of water movement. Therefore, the biomass in each modelled size class of fish is not modelled as a passive tracer subject to hydrodynamic transport (e.g., turbulent mixing). Instead, fish biomass per size class is described by depth-integrated densities (g wet mass $m^{-2}$). This is different from the planktonic prey that the smallest sizes of fish feed on; plankton variables in the LTL model are treated as 3D tracers subject to advection and diffusion. As a result, a key challenge in coupling FABM-MIZER to LTL models is the change between a 3D (depth, x,y) system in the LTL and 2D (x,y) system for FABM-MIZER. In order to enable two-way coupling, the vertical distribution of fish had to be assumed so that a) perceived prey can be calculated and consequently decreased in the LTL, and b) that waste (produced by excretion and respiration) can be added back into the LTL model.

Following Cheung et al. (2018) it is assumed that fish distribute instantaneously in the vertical according to food availability. The vertical distribution of fish, f(z), is thus proportional to the total concentration (mg carbon $m^{-3}$) of plankton prey, $P_T(z)$:

$$f(z) = \frac{P_T(z)}{\int P_T(z)dz} \qquad (S3)$$

where z is the depth in m. This is a probability density with a depth integral of 1 and units of $m^{-1}$. The same distribution is assumed for all size classes of fish. The total concentration of prey itself is the sum of the concentrations of ERSEM's individual phytoplankton functional types (PFTs), Pj(z):

$$P_T(z) = \sum_{j=1}^{7} P_j(z) \qquad (S4)$$

An overview of ERSEM's 7 pelagic PFTs and their corresponding sizes is given in Table 1.

As fish are assumed to distribute themselves rapidly in the vertical dimension, the distribution $f(z)$ can be interpreted as a measure of the time spent by an individual fish at depth $z$. For any depth-dependent environmental variable, $c(z)$, such as temperature, the effective experienced value is thus:

$$\int f(z)c(z)dz \qquad (S5)$$

(Note, the required normalisation factor $\int f(z)dz$ equals 1 and can therefore be omitted). This

formulation implies that if fish are distributed in a very thin layer (e.g., all near the surface), their effective environment is of that layer only. If fish are distributed homogeneously throughout the water column, the effective environment is the mean of conditions across the entire column. Similar to the treatment of environmental variables, the vertical distribution of waste production (matter $m^{-3}$ $time^{-1}$) is the product of depth-integrated waste production (matter $m^{-2}$ $time^{-1}$) and the vertical distribution $f(z)$. The workflow for converting between 3D and 2D model domains and 2-way coupling is shown in Figure 1 in the main text.

### *S1.3 Prey encounter and ingestion*

At any depth, the local ingestion rate, $I_i$, for a single predator (in units of prey biomass $time^{-1}$) in size class $i$ is

$$I_i(z) = I_{max} \frac{E_i}{E_i + I_{max}} \quad (S6)$$

Where $I_{max}$, is the specific maximum ingestion rate and $E_i$ is the encounter rate for a predator from size class $i$. When $I_{max}$ is large, the ingestion rate becomes the encounter rate. The encounter rate, $E_i$, is given by

$$E_i(z) = V_i \sum_{j=1}^{n_p} \phi_{ij} p_j\,(z) \quad (S7)$$

where $V_i$ is the predator's volumetric search rate ($m^3$ $time^{-1}$), $n_p$ the number of prey types (ERSEM plankton functional types (*nprey*) and all size classes of fish (*nclass*)), $p_j(z)$ the local concentration of the biomass of prey $j$ and $\phi_{ij}$ the preference of a predator from size class $i$ for prey $j$.

Following Blanchard et al. 2009, the volumetric search rate, $V_i$, is calculated as:

$$V_i = \gamma {w_i}^{q-1} \quad (S8)$$

Where γ is the volumetric search rate when $w$=1g and $q$ is the exponent of the scaling of volumetric search rate with mass. Similarly, the specific maximum ingestion rate, $I_{max}$, is given by

$$I_{max} = h {w_i}^{n-1} \quad (S9)$$

where *h* is the maximum ingestion rate when *w*=1g and *n* is the exponent of the scaling of maximum ingestion rate with mass.

Predators feed preferentially on prey which are small enough to be ingested but large enough to maximise energy intake (Cohen et al., 1993). The preference of a predator with mass *w* for a prey of mass $w_p$ is a log-normal distribution of the predator : prey mass ratio:

$$\phi(w, w_p) = \exp\left[\frac{-\left(\ln\frac{\beta w_p}{w}\right)^2}{2\sigma^2}\right] \qquad (S10)$$

where σ is the width of the distribution of prey size preference and $\beta$ is the optimal predator-to-prey mass ratio. The preference function has a maximum of 1, which occurs when the predator mass is equal to the prey mass multiplied by beta.

For a prey $j$ that is assigned a single mass $w_j$ (e.g., each size class of fish), equation S11 fully specifies the preference of a predator $i$ for such prey:

$$\phi_{ij} = \phi(w_i, w_j) \qquad (S11)$$

However, ERSEM prey types are assigned a mass range between $w_{j,min}$ and $w_{j,max}$ rather than a single mass. To compute the effective preference for such prey, we assume a uniform distribution of prey concentration as function of log prey size within this window; that is equivalent to a Sheldon type spectrum for that prey (Sheldon et al., 1972). Under this assumption, the effective preference of predator $i$ for prey $j$ is the average preference over the prey's mass range:

$$\phi_{ij} = \frac{1}{\ln w_{j,max} - \ln w_{j,min}} \int_{\ln j,min}^{\ln j,max} \phi(w_i, e^{\omega})\, d\omega \qquad (S12)$$

All preferences are independent of prey concentration and can therefore be computed at the start of the simulation from known mass (or mass range) of prey and predator.

The depth-integrated ingestion rate (prey $m^{-2}$ $time^{-1}$) per predator in size class $i$ is the integral

of local ingestion rates $I_i$ (eq 6)

$$I_{Ti} = \int I_i dz = \int V_i \sum_{j=1}^{n_p} \phi_{ij} p_j(z) f(z) dz = V_i \sum_{j=1}^{n_p} \phi_{ij} \int p_j(z) f(z) dz \quad (S13)$$

Here, $\int p_j(z) f(z) dz$ is the depth-averaged concentration of prey $j$, weighted by the vertical distribution of the predator, $f(z)$. For ERSEM prey types, the effective prey concentration experienced by the predator is computed directly from $p_j(z)$ and $f(z)$. For prey that are fish from other size classes, the depth-explicit concentration $p_j(z)$ is the product of the depth-integrated density $W_j$ and $f(z)$, which means that the depth-averaged concentration experienced by the predator is:

$$\int p_j(z) f(z) dz = W_j \int [f(z)]^2 dz \quad (S14)$$

The preference function $f(z)$ appears twice, because it applies both to predator and prey. The resulting factor $\int [f(z)]^2 dz$ has dimension $m^{-1}$ and can be thought of as the reciprocal of an "interaction depth": the effective depth range over which the fish interact. All predator-prey interactions within the fish community are inversely proportional to this depth.

By substituting prey carbon, nitrogen, phosphorus and silicon (diatoms only) for prey $p_j$ in the above, we obtain separate depth-integrated ingestion rates $I_{Ci}$ for carbon (mmol C $time^{-1}$), $I_{Ni}$ for nitrogen (mmol N $time^{-1}$), $I_{Pi}$ for phosphorus (mmol P $time^{-1}$) and $I_{Si}$ for silicon (mmol Si $time^{-1}$) per predator. Note that as silicon is not part of fish biomass, its ingestion is tracked only to allow the same quantity to be egested immediately, thus ensuring Si conservation.

Only a fraction, $\alpha$, of ingested prey is assimilated. This fraction is assumed to be the same for C, N and P. The unassimilated remainder leaves the fish through egestion of particulate matter (faeces) or respiration/excretion of dissolved matter. These two fluxes contribute to different ERSEM pools and are therefore tracked separately. Following Blanchard et al. (2009) we assume a constant fraction $\alpha_{eg}$ of ingested food is egested. That leaves a fraction $1 - \alpha - \alpha_{eg}$ to be excreted as phosphate and ammonium or respired as carbon dioxide.

The specific maintenance rate ($s^{-1}$) of fish is described as:

$$kw = ks.w_i^p \quad (S15)$$

where ks is the standard metabolism when w=1g, and p is exponent of standard metabolism.

Any maintenance costs for standard metabolism (not active in community mode, and thus this study) are assumed to be energetic costs paid by breaking down carbon (carbohydrates or lipids). Such costs ($kw$ ; g time$^{-1}$) are therefore subtracted from the incoming carbon flux only; they do not diminish the availability of nitrogen or phosphorus. The final incoming fluxes of carbon, nitrogen and phosphorus are converted to the biomass they can produce; the final expression for somatic growth (wet mass in g) per predator is determined by the minimum of the three fluxes:

$$g_i = 0.12 min(\alpha I_{Ci} - kw, \alpha I_{Ni}/q_{NC}, \alpha I_{Pi}/q_{PC}) \quad (S16)$$

where 0.12 is the wet mass to carbon ratio (g per mmol), and $q_{NC} = 16/106$ and $q_{PC} = 1/106$ is the Redfieldian stoichiometric ratios of nitrogen and phosphorus to carbon. These default ratios are based on conventions in marine biogeochemical, modelling. While they fall within the range of ratios observed by (Sterner and George, 2000), they could be modified in the future to account for the tendency of higher trophic levels of be richer in phosphorus (Le Mézo et al., 2022; Sterner and Elser, 2002). If ingested carbon does not suffice to pay maintenance, a starvation mortality is introduced as in Blanchard et al. (2014) and somatic growth is set to 0.

Typically, assimilated fluxes of carbon, nitrogen and phosphorus ($I_{Ci}, I_{Ni}, I_{Pi}$) will not come in the ratio needed for synthesis of fish biomass. With the somatic growth rate $g_i$ set by the most limiting element, excess assimilated fluxes of carbon, nitrogen and phosphorus amount to $(\alpha I_{Ci} - g_i/0.12)$, $(\alpha I_{Ni} - q_{NC}\, g_i/0.12)$ and $(\alpha I_{Pi} - q_{PC}\, g_i/0.12)$ respectively. These excess fluxes are egested and contribute to the detritus pools tracked by the LTL, in this case the large size class of detritus within ERSEM.

### *S1.4 Recruitment*

As we are describing a community of fish, rather than a single species, there is no straightforward relationship between egg production and the biomass in larger ("adult") size classes – reproductive effort per size class would depend on species composition, which is unknown. Instead of explicitly modelling reproduction, egg production is therefore dictated by the presence of plankton prey. Specifically, the density of the smallest size class of fish (1 mg) is relaxed towards an expected density that is obtained by extrapolating the depth-integrated

plankton size spectrum. The scale factor and exponent of this spectrum are computed at run time by linear regression.

To obtain the plankton size spectrum, the depth-integrated wet mass of each plankton prey is first computed, by depth-integrating its carbon concentration (mmol C $m^{-3}$) and multiplying with 0.12 g wet mass per mmol C. A discretised plankton size spectrum along a log-spaced size axis is constructed, spanning $1/\beta^2$ to 1 mg in wet mass with a total of 100 bins. For each bin $k$, the natural logarithm of wet mass in the centre is denoted by $\omega_k$. The density of each plankton type is assumed to be uniformly distributed (Sheldon et al., 1972) between their predefined size ranges (Table 1 in the main text), which means the final density spectrum (g $m^{-2}$) is a sum of multiple uniform distributions. For each bin $k$, the biomass density is divided by the bin width (in g wet mass) to obtain a representation of the spectrum in $m^{-2}$ that is independent of its discretisation; finally we then take its natural logarithm, denoted as $y_k$. The spectral slope (= exponent on linear scale) through linear regression is estimated as

$$a = \frac{\frac{1}{n}\sum \omega_k y_k - \acute{\omega}\acute{y}}{\frac{1}{n}\sum {\omega_k}^2 - \acute{\omega}^2} \quad where \quad \acute{\omega} = \frac{1}{n}\sum \omega_k \text{ and } \acute{y} = \frac{1}{n}\sum y_k \qquad (S17)$$

The spectral offset (= log of the scale factor on linear scale) is subsequently given by

$$b = \frac{1}{n}\sum y_k - a\omega_k \qquad (S18)$$

The expected density of the smallest size class of fish (1 mg in wet mass), in g $m^{-2}$, is then estimated as $\exp[a\ln 0.001 + b]$, multiplied by the bin width of that smallest size class (in g wet mass).

If this expected density of the smallest size class is greater than the current density, the latter is relaxed towards the expected value at a rate of 5 $d^{-1}$. If the expected density is smaller, relaxation is not applied. In this manner, we ensure that effective "recruitment" is always positive. In order to achieve mass balance, any excess/supply of C/P/N needed for recruitment is returned/taken from the POM pool, in this case large POM.

### *S1.5 Fishing*

As in Scott et al. (2014), the implementation supports several fishing parameterisations, including "knife edge" (a constant mortality applied to all fish above a user-selected wet

mass), logistic selectivity (Blanchard et al., 2014), or a mortality that increases linearly with the logarithm of wet mass (Blanchard et al., 2009). For the community configuration used in this study, we follow Blanchard et al. (2012) and Scott et al. (2014) in prescribing knife-edge fishing mortality ($F_m$) where a constant rate of fishing occurs ($F$, $yr^{-1}$) for all fish over a certain size ($w_f$) on the entire domain:

$$F_m = F \; if \;\; w_i > w_f \; else \; 0 \tag{S19}$$

Biomass removed through application of the fisheries mortality is tracked as "landings" that no longer interact with the ecosystem. Effectively, this reduces the quantities of bioavailable carbon, nitrogen and phosphorus, which if left unchecked would lead to long-term drift of the ecosystem state. Such drift is not seen in reality, likely because part of the fished biomass immediately re-enters the system as discards, while loss of the remaining part is compensated by non-local nutrient sources (e.g., riverine inputs).

### *S1.6 Intrinsic mortality*

Intrinsic mortality combines a background mortality that is either constant or an allometric function of individual mass, and a senescence mortality that is formulated as allometric function of size

$$\mu_i = \mu_b {w_i}^{zb} + \mu_s \left(\frac{w_i}{w_s}\right)^{z_s} \tag{S20}$$

where $\mu_b$ and $\mu_s$ are the prefactor for background and senescence mortality, respectively, $z_b$ and $Z_s$ are the exponent of scaling for the background and senescence mortality rate with mass and $w_s$ is the mass at which senescence mortality precisely equal $\mu_s$ and in which senescence mortality starts to become more important. The first term corresponds to the background mortality, and affects smaller individuals more, and the second term corresponds to senescence mortality and predominately affects larger individuals.

### *S1.7 Temperature dependence*

The default parametrisation for temperature dependence is given for an ambient temperature of 13°C. All rates are then scaled with the dimensionless Arrhenius factor:

$$\tau = \exp\left(c_1 - \frac{E_a}{k_B T}\right) \tag{S21}$$

With $E_a$ representing activation energy, $k_B$ the Boltzmann constant ($8.62 \times 10^{-5}$ eV $K^{-1}$) and $T$ the ambient temperature in Kelvin. Constant $c_1$ equals $\frac{E_a}{k_B T}$ for reference temperature 286.15 K (13°C).

As in Blanchard et al., (2012) this temperature dependence is applied to predator-prey interactions (thus growth and predation mortality vary with temperature) and background mortality, but not senescence mortality. Fishing mortality is temperature independent, as it is a prescribed external pressure.

**TABLES**

| Parameter | | Value | Units | Reference |
|---|---|---|---|---|
| *w_min* | Egg mass | 1e-3 | g | a,b,d |
| *w_inf* | Maximum fish size class | 1e6 | g | a,b,d |
| *nclass* | Number of fish size classes | 100 | - | d |
| *nprey* | Number of planktonic prey | 7 | - | |
| Tref | Reference Temperature | 13 | $^{o}C$ | |
| Ea | activation energy | 0.63 | eV | ab |
| $c_1$ | Constant in Arrhenius function | 25.55 | - | b |
| $\beta$ | Optimal predator: prey mass ration | 100 | - | abcde |
| $\sigma$ | width of prey size preference | 2.3025 | - | ab |
| $\gamma$ | Pre-factor for volumetric search rate | 91.32 | $m^3\ yr^{-1}\ g^{-q}$ | f |
| $q$ | exponent of search volume | 0.776 | - | f |
| $\alpha$ | Assimilation efficiency | 0.2 | - | abd |
| $\alpha_{eg}$ | Fraction of food that is egested | 0.3 | - | a |
| $n$ | Exponent of maximum consumption | 2/3 | - | c,d |
| $Z_b$ | Exponent of background mortality | -0.25 | - | a,b |
| $\mu_b$ | Pre-factor for background mortality | 0.2 | $yr^{-1}$ | a |
| *W_s* | start mass for senescence mortality | 1000 | g | b |
| $Z_s$ | exponent for senescence mortality | 0.3 | - | a,b |
| $\mu s$ | pre-factor for senescence mortality | 0.2 | $yr^{-1}$ | a,b,d |
| *ks* | pre-factor for standard metabolism | 0 | $yr^{-1}\ g^{-p}$ | d |
| $p$ | Exponent of standard metabolism | 0.7 | - | c |
| $h$ | Pre-factor for maximum ingestion rate | 1e9 | $yr^{-1}\ g^{-n}$ | |
| *R_relax* | relaxation towards expected egg density | 5 | $d^{-1}$ | |
| $w_f$ | minimum mass for fishing selectivity | 80 | g | |
| $F$ | Fishing effort | 0.8 | $yr^{-1}$ | b |

**Table S1:** Parameters for FABM-MIZER presented in this study. Parameters can be matched to equations in the text. References: a Blanchard et al., 2009, b Blanchard et al., 2012, c Blanchard et al., 2014, d Scott et al., 2014, e Hartvig et al., 2011, f Acuña et al., 2011

| Reference | **q** | **γ** |
|---|---|---|
| Blanchard (2009) | 0.9262 | 156 |
| Blanchard (2012) | 0.82 | 640 |
| Kiørboe and Hirst (2014) | 0.76 | 410 |
| Acuña et al. (2011) (unmodified) | 0.926 | 156 |
| This study | 0.776 | 91.32 |

**Table S2:** Parameters for q in gamma used in equation S8 and to calculate lines in Figure S2

| Test | $W_f$ | $F$ | Greater North Sea | Celtic Sea | Whole Domain |
|---|---|---|---|---|---|
| Observed | - | - | 1.75* | 0.76** | - |
| 1 | 1.25 | 0.8 | 3.53 | 3.03 | 2.80 |
| 2 | 1.25 | 0.6 | 1.99 | 1.55 | 1.40 |
| 3 | 80 | 0.8 | 1.05 | 0.61 | 0.58 |
| 4 | 80 | 0.6 | 0.90 | 0.54 | 0.51 |

**Table S3:** Calibration tests for fish landed (g Wm $m^{-2}$ $yr^{-1}$) in ICES Greater North Sea and ICES Celtic Sea regions. $W_f$ and F represent size above which fish are fished and fishing mortality respectively (equation S19). The model was initially run for 5 years from 1981-1985, with results for 1985 with the exception of test1 where model was run for 2000-2002 and results taken for 2002. *1985-2017; ICES 2021b, ** 1985-2917 ICES 2021a

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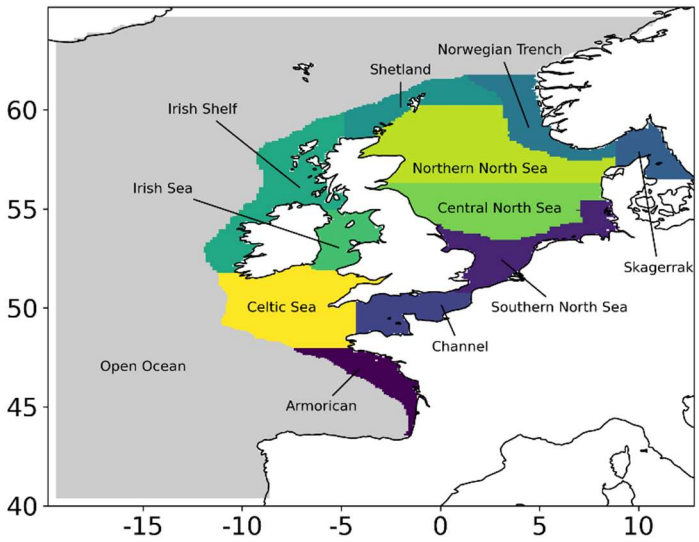



**Figure S1:** Regions of shelf used in this paper following Wakelin et al. (2012). Grey area represents Open Ocean

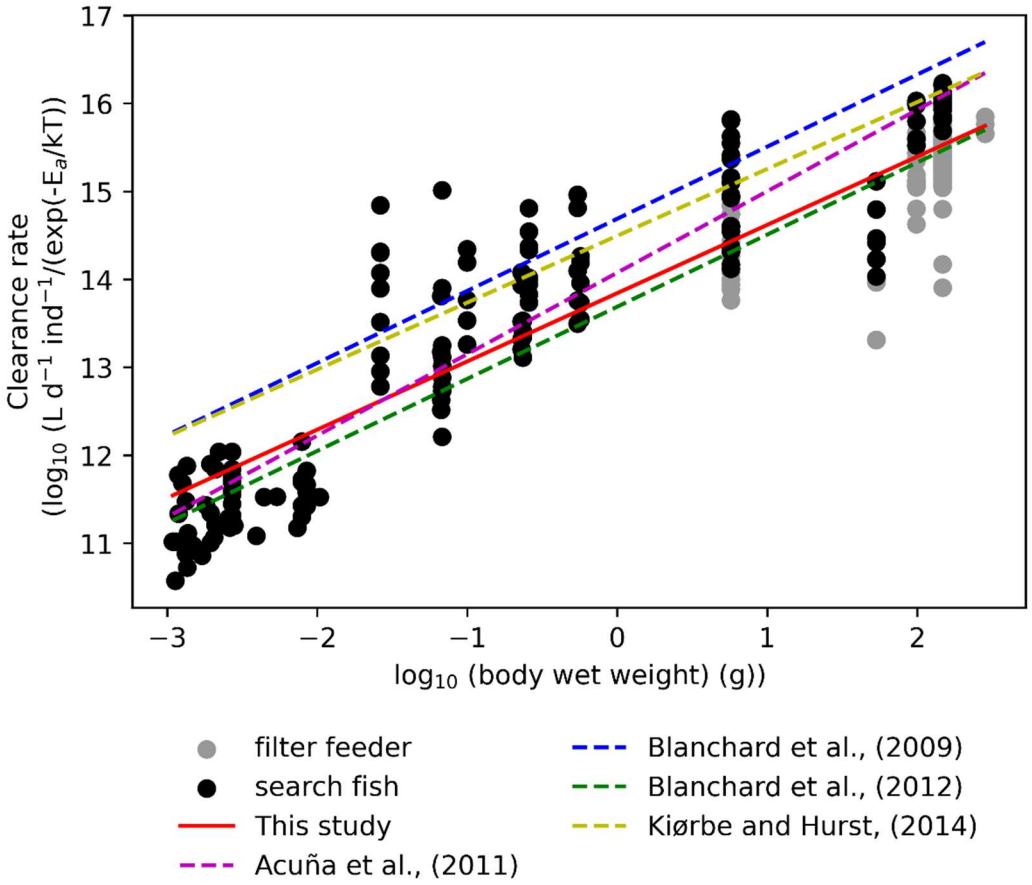



**Figure S2:** Comparison of temperature-normalised clearance rates calculated from data (grey and black dots; Acuña et al. (2011)) and using parameters from other literature. Gamma and q (equation S8) were calculated from intercept and slope respectively with gamma corrected for the temperature equation used FABM-MIZER (equation S21).The Acuna et al. line uses the gamma and q reported in their manuscript (only search fish) whereas our study uses all fish and limits the data to above $10^{-3}$ g wet weight. Gamma and q for each line are reported in Table S2.

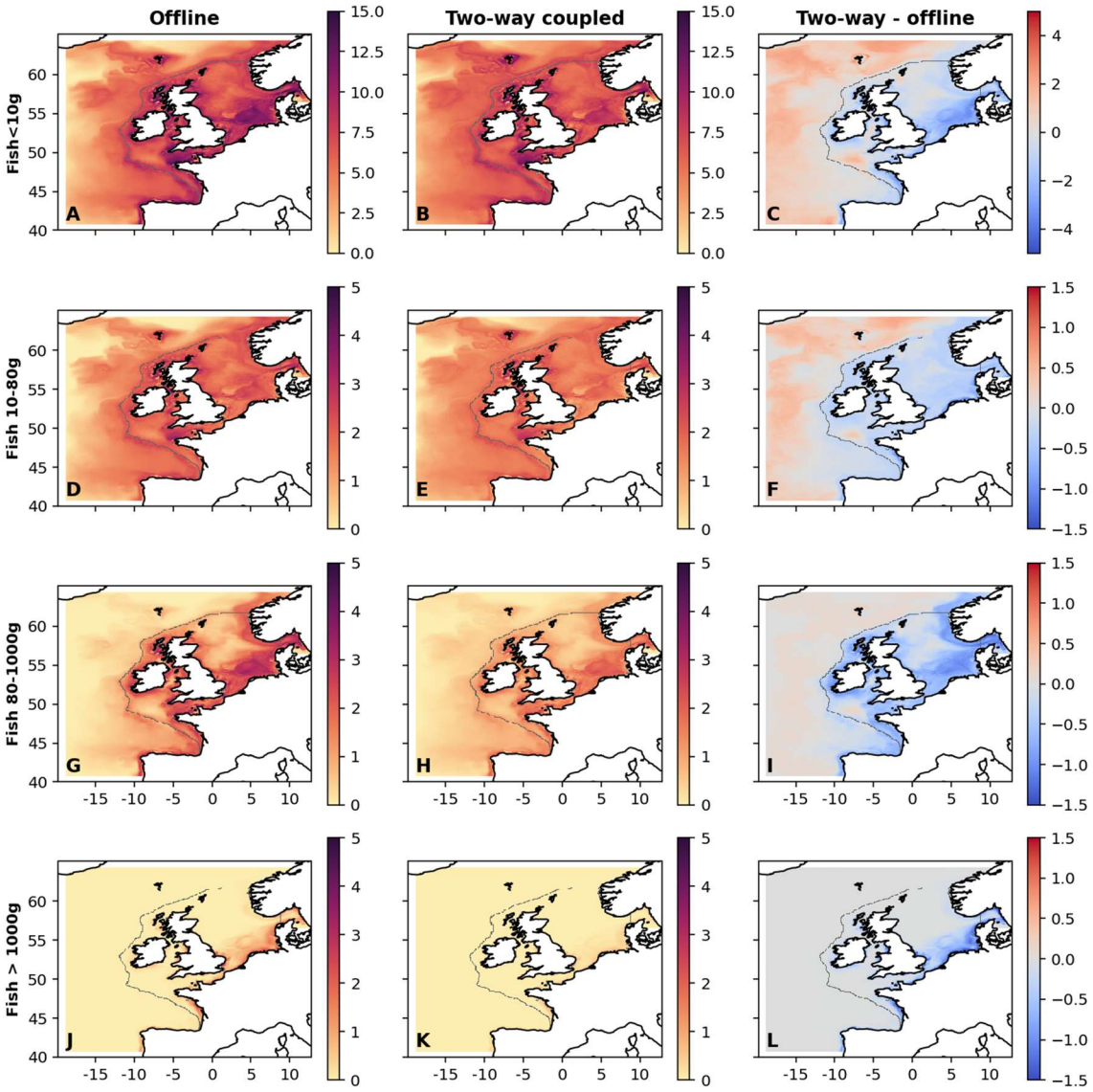



**Figure S3:** Biomass (gWM $m^{-2)}$ of size classes of fish across the NW European Shelf between offline and two-way coupled model. In the difference plots, blue/red indicates the two-way coupled model is lower/greater than the offline model. Note difference in scale for <10g fish and all other plots. Grey line depicts the edge of the North West European Shelf.

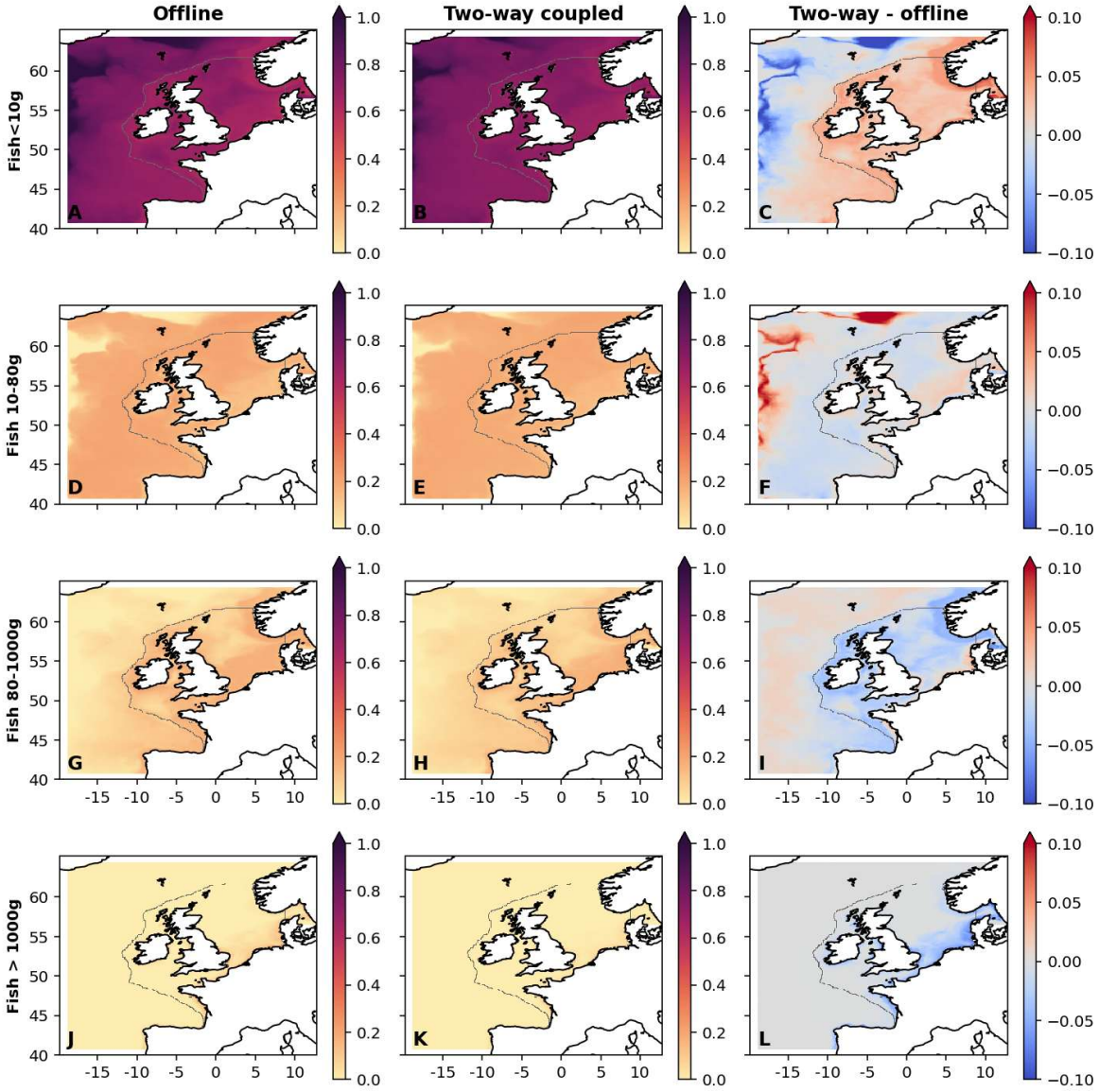



**Figure S4:** Fraction of the total biomass of different size classes of fish across the NW European Shelf in the offline (1st column) and two-way coupled (2nd column) models and difference between them (3rd column). In the difference plots, blue/red indicates the two-way coupled model is lower/greater than the offline model. Note difference in scale for <10g fish and all other plots. Grey line depicts the edge of the North West European Shelf.

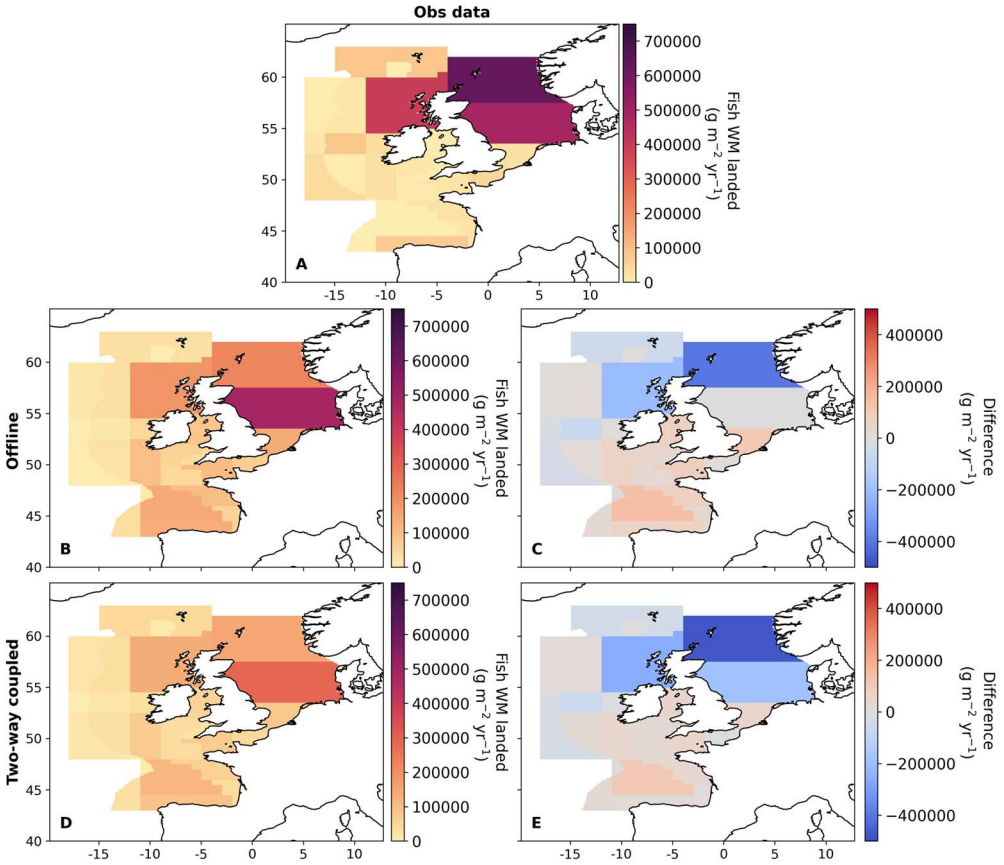



**Figure S5:** Validation of (A) ICES catch data for pelagic fish (ICES, 2023) with (B) offline and (D) two-way coupled model predictions for landing data from 2006-2017. Model results were summed over the ICES areas. Panels (C) and (E) show differences between model simulation and observational data for the offline and two-way coupled models respectively. Blue/red indicates fish catches were lower/greater in the offline/two-way coupled simulation than the observational data.

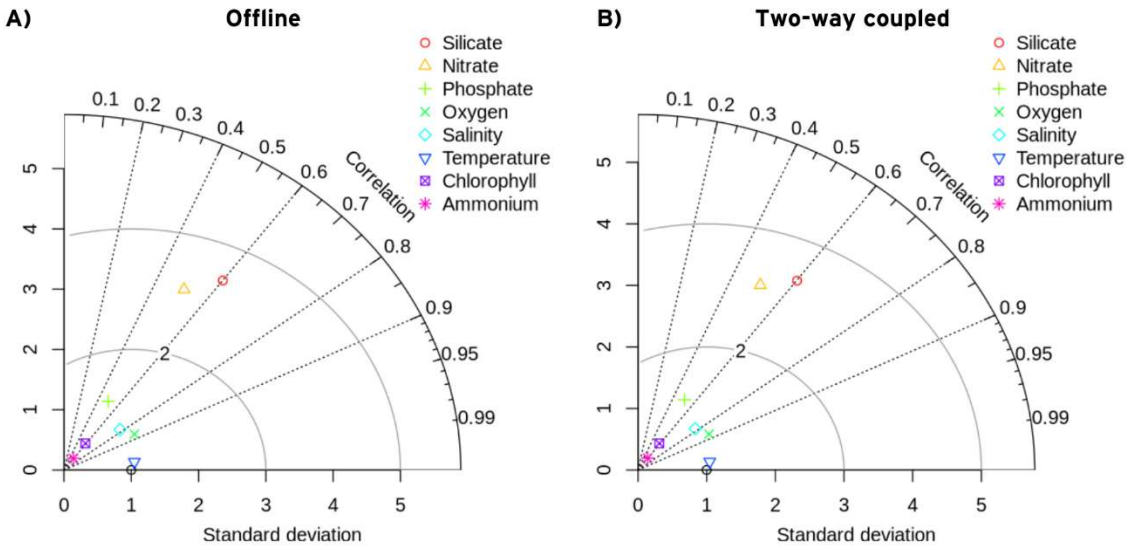



**Figure S6**: Taylor diagram for surface annual mean of ammonium, chlorophyll, nitrate, oxygen, phosphate, salinity, silicate, and temperature. Climatological annual averages of the model between 1986-2017 are compared against gridded observations across the model's spatial domain. Standard deviation is normalized by the standard deviation of the observations, and a standard deviation below 1 indicates that the model is less variable than the observations. Note: This figure summarizes the overall ability of the model to reproduce climatological spatial patterns, and it does not represent temporal performance. All observational data except temperature was taken from the North Sea Biogeochemical Climatology dataset database (Hinrichs et al., 2017) whilst temperature was taken from OSTIA dataset (CMEMS, 2023)

S1.

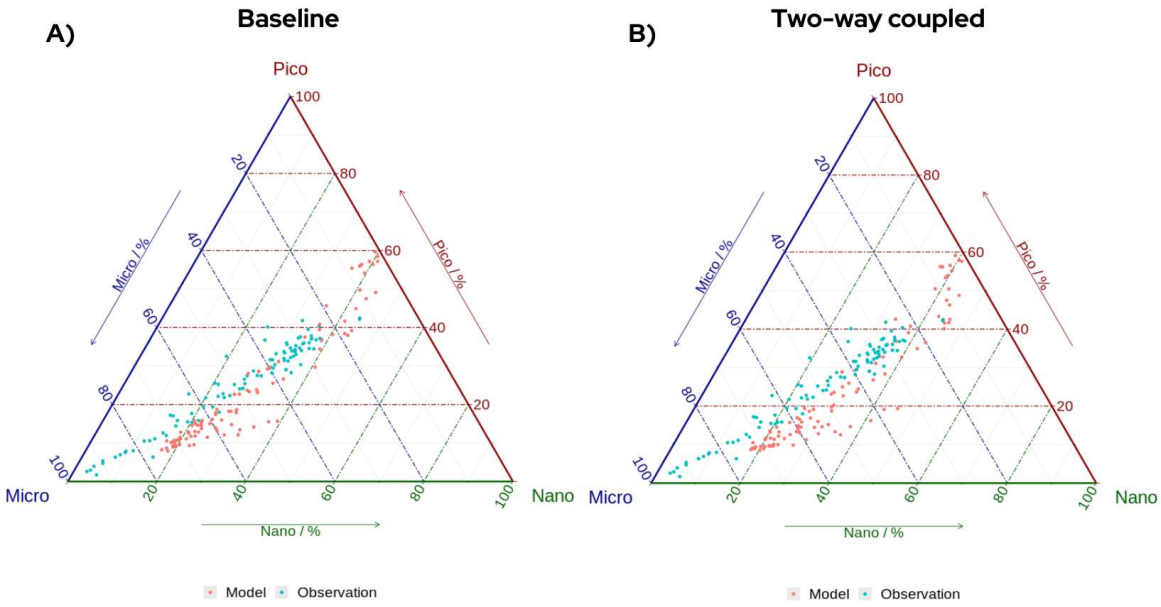



**Figure S7:** Ternary plot comparing surface (top 5m) phytoplankton functional types in the a) baseline and B) two-way coupled simulations against observational results (Creach and Forster, 2017) across the North Sea for August/September 2010 and 2011.

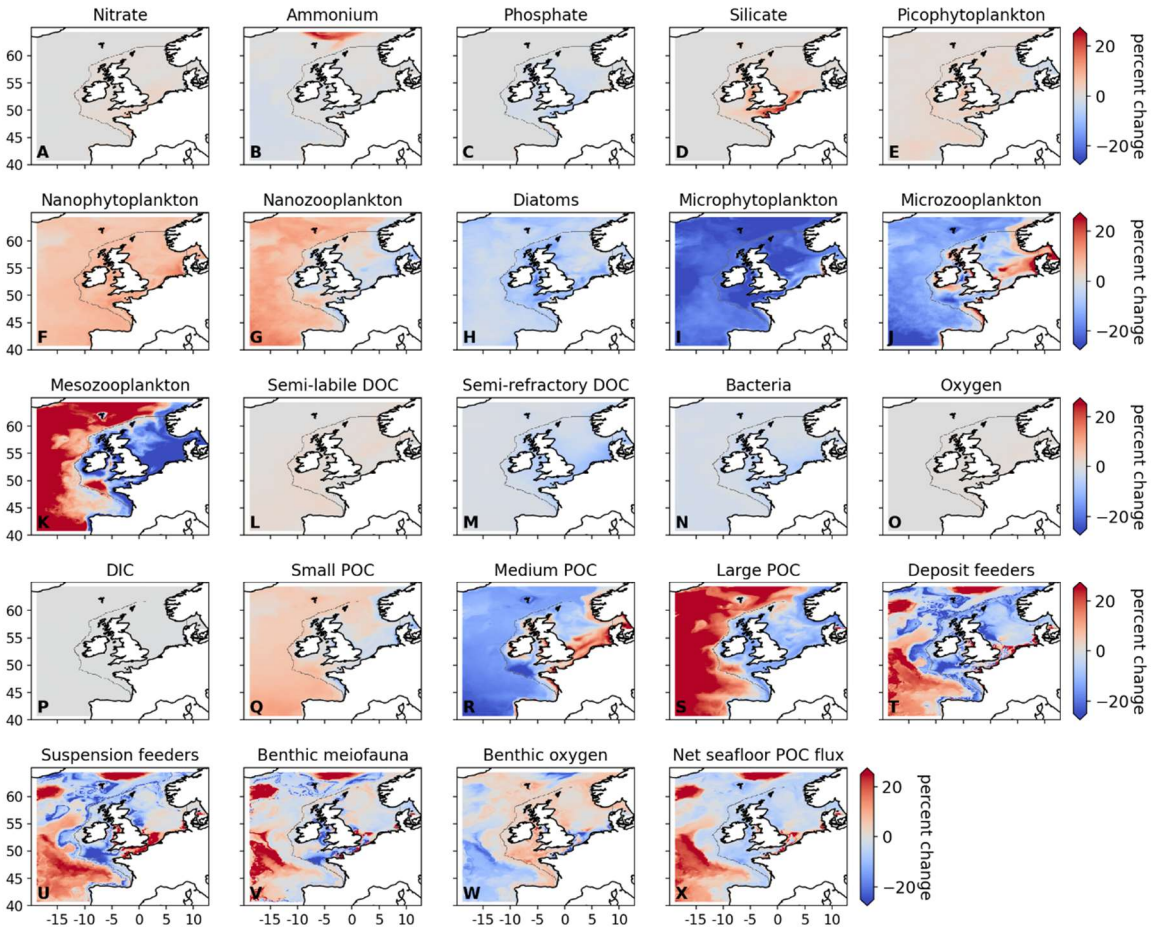



**Figure S8:** Percent difference in lower trophic level pools using annual mean outputs between the two-way coupled simulation and baseline simulation over 1986-2017. Dissolved constituents show changes in the water column mean, while particulate pools (plankton, bacteria and POC) show changes in the depth integrated values. Blue/red indicates values were lower/greater in the two-way coupled simulation than baseline simulation

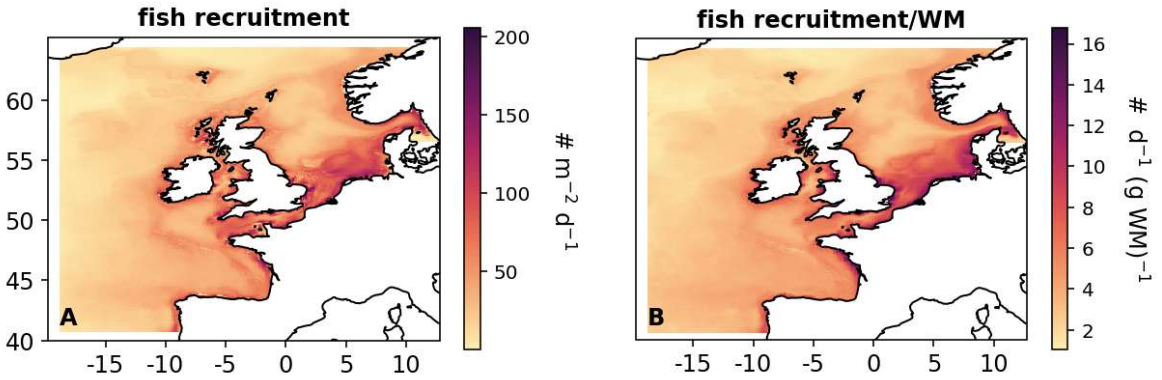



**Figure S9:** (A) Fish recruitment and (B) ratio of fish recruitment to wet mass averaged over 1986-2017 for the two-way coupled simulation

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